INFO-I590 Fundamentals and Applications of LLMs

DPO and GRPO

Direct Preference Optimization (DPO)
Group Relative Policy Optimization (GRPO)

Jisun An

Direct Preference Optimization (DPO)

RLHF optimization objective

Maximizes the rewards

Prevents the model from changing too drastically

$$\max_{\pi} \mathbb{E}_{x \sim D, y \sim \pi} \left[r(x, y) - \beta D_{KL}(\pi(y \mid x) \parallel \pi_{\text{ref}}(y \mid x)) \right]$$

Current aligned LLM

SFT (instruction-tuned) LLM

How to get rid of this reward?
(not get rid of it, but turn it into a probability)

Can we avoid rollouts?

DPO (Direct Preference Optimization)

- Idea: Can we fine-tune a model directly using a preference dataset without RL?
- Why Avoid RL?
  - RL is unstable and computationally expensive.
  - It’s hard to pinpoint what to change in long outputs.
  - Generating multiple samples for training (rollouts) increases cost.
  - Training a reliable reward model is challenging.
  - Small reward differences can cause instability.
- Also called as ‘Preference Tuning’

The DPO loss function

Bradley Terry Model

$$\max_{\pi} \mathbb{E}_{x\sim D, y\sim\pi}[r(x,y) - \beta D_{KL}(\pi(y \mid x) \parallel \pi_{\text{ref}}(y \mid x))]$$

$$P(y_W > y_L \mid x) = \frac{e^{r(x,y_W)}}{e^{r(x,y_W)} + e^{r(x,y_L)}}$$

Mathematical Manipulation

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right]$$

The DPO loss function

Maximize probability for good response

Minimize probability for bad response

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right]$$

Aligned model we are training

Don't change the model too much

- No explicit reward model
- No need for rollouts from the policy

Rafailov et al. 2023. Direct preference optimization: Your language model is secretly a reward model. https://arxiv.org/abs/2305.18290

(Optional)

Deriving the DPO loss function

$$
\begin{aligned}
&\max_{\pi} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi}\left[r(x,y)\right] - \beta\mathbb{D}_{\mathrm{KL}}\left[\pi(y|x) \mid\mid \pi_{\mathrm{ref}}(y|x)\right] \\
&\qquad = \max_{\pi} \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y\sim\pi(y|x)}\left[r(x,y) - \beta\log\frac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)}\right] \\
&\qquad = \min_{\pi} \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y\sim\pi(y|x)}\left[\log\frac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)} - \frac{1}{\beta}r(x,y)\right] \\
&\qquad = \min_{\pi} \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y\sim\pi(y|x)}\left[\log\frac{\pi(y|x)}{\frac{1}{Z(x)}\pi_{\mathrm{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right)} - \log Z(x)\right] \qquad (12)
\end{aligned}
$$

where we have partition function:

$$
Z(x) = \sum_{y} \pi_{\mathrm{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right).
$$

* Note that the partition function is a function of only x and the reference policy π_{ref}, but does not depend on the policy π.

Rafailov et al. 2023. Direct preference optimization: Your language model is secretly a reward model. https://arxiv.org/abs/2305.18290

(Optional)

We can now define a new policy π*,

$$\pi^*(y|x) = \frac{1}{Z(x)}\pi_{\text{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right),$$

which is a valid probability distribution as $\pi^*(y|x) \geq 0$ for all y and $\sum_y \pi^*(y|x) = 1$. Since $Z(x)$ is not a function of y, we can then re-organize the final objective in Eq 12 as:

$$\min_\pi \mathbb{E}_{x\sim\mathcal{D}}\left[\mathbb{E}_{y\sim\pi(y|x)}\left[\log\frac{\pi(y|x)}{\pi^*(y|x)}\right] - \log Z(x)\right] = \tag{13}$$

$$\min_\pi \mathbb{E}_{x\sim\mathcal{D}}\left[\mathbb{D}_{\text{KL}}(\pi(y|x)\ ||\ \pi^*(y|x)) - \log Z(x)\right] \tag{14}$$

Now, since $Z(x)$ does not depend on π, the minimum is achieved by the policy that minimizes the first KL term. Gibbs' inequality tells us that the KL-divergence is minimized at 0 if and only if the two distributions are identical. Hence we have the optimal solution:

$$\pi(y|x) = \pi^*(y|x) = \frac{1}{Z(x)}\pi_{\text{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right) \tag{15}$$

for all $x \in \mathcal{D}$. This completes the derivation.

Rafailov et al. 2023. Direct preference optimization: Your language model is secretly a reward model. https://arxiv.org/abs/2305.18290

(Optional)

$$\pi_r(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

We can rearrange the above equation to express the reward function in terms of its corresponding optimal policy π_r, the reference policy π_{ref}, and the unknown partition function(x).

$$r(x, y) = \beta \log \frac{\pi_r(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x)$$

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(Optional)

It is straightforward to derive the DPO objective under the Bradley-Terry preference model as we have

$$p^*(y_1 \succ y_2|x) = \frac{\exp\left(r^*(x, y_1)\right)}{\exp\left(r^*(x, y_1)\right) + \exp\left(r^*(x, y_2)\right)} \tag{16}$$

In Section 4 we showed that we can express the (unavailable) ground-truth reward through its corresponding optimal policy:

$$r^*(x, y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x) \tag{17}$$

Substituting Eq. 17 into Eq. 16 we obtain:

$$\begin{aligned} p^*(y_1 \succ y_2|x) &= \frac{\exp\left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)} + \beta \log Z(x)\right)}{\exp\left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)} + \beta \log Z(x)\right) + \exp\left(\beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)} + \beta \log Z(x)\right)} \\ &= \frac{1}{1 + \exp\left(\beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)} - \beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)}\right)} \\ &= \sigma\left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)} - \beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)}\right). \end{aligned}$$

Rafailov et al. 2023. Direct preference optimization: Your language model is secretly a reward model. https://arxiv.org/abs/2305.18290

(Optional)

Now that we have the probability of human preference data in terms of the optimal policy rather than the reward model, we can formulate a maximum likelihood objective for a parametrized policy π_θ. Analogous to the reward modeling approach (i.e. Eq. 2), our policy objective becomes:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right]. \tag{7}$$

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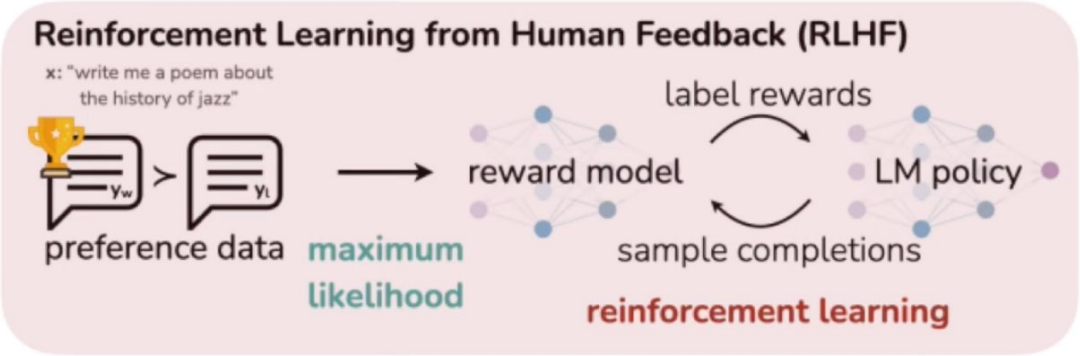
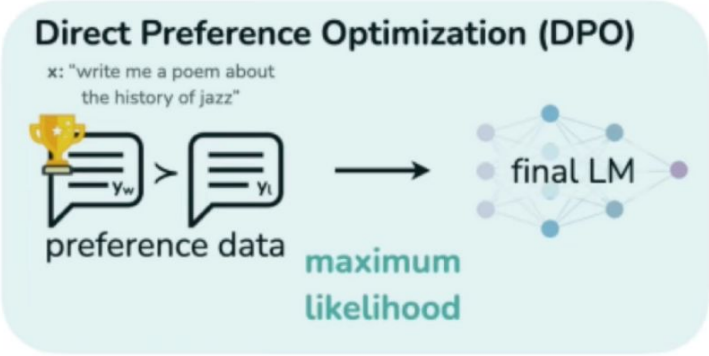
RLHF vs DPO





Img: https://www.youtube.com/watch?v=k2pD3k1485A

Group Relative Policy Optimization (GRPO)

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

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Abstract

We introduce our first-generation reasoning models, DeepSeek-R1-Zero and DeepSeek-R1. DeepSeek-R1-Zero, a model trained via large-scale reinforcement learning (RL) without supervised fine-tuning (SFT) as a preliminary step, demonstrates remarkable reasoning capabilities. Through RL, DeepSeek-R1-Zero naturally emerges with numerous powerful and intriguing reasoning behaviors. However, it encounters challenges such as poor readability, and language mixing. To address these issues and further enhance reasoning performance, we introduce DeepSeek-R1, which incorporates multi-stage training and cold-start data before RL. DeepSeek-R1 achieves performance comparable to OpenAI-o1-1217 on reasoning tasks. To support the research community, we open-source DeepSeek-R1-Zero, DeepSeek-R1, and six dense models (1.5B, 7B, 8B, 14B, 32B, 70B) distilled from DeepSeek-R1 based on Qwen and Llama.

DeepSeek-AI 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. https://arxiv.org/abs/2501.12948

2024 AIME I Problems

Problem 1

Every morning Aya goes for a 9-kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks at $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.

Solution

Problem 2

There exist real numbers x and y, both greater than 1, such that $\log_x(y^x)=\log_y(x^{4y})=10$. Find xy.

Solution

Problem 3

Alice and Bob play the following game. A stack of n tokens lies before them. The players take turns with Alice going first. On each turn, the player removes 1 token or 4 tokens from the stack. The player who removes the last token wins. Find the number of positive integers n less than or equal to 2024 such that there is a strategy that guarantees that Bob wins, regardless of Alice's moves.

Solution

Problem 4

Jen enters a lottery by selecting 4 distinct elements of $S=\{1,2,3,\cdots,9,10\}$. Then four elements of S are drawn at random. Jen wins a prize if at least two of her numbers were drawn, and wins the grand prize if all four of her numbers are drawn. The probability that Jen wins the grand prize given that Jen wins a prize is $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m+n$.

Solution

Img: https://artofproblemsolving.com/wiki/index.php/2024_AIME_I_Problems

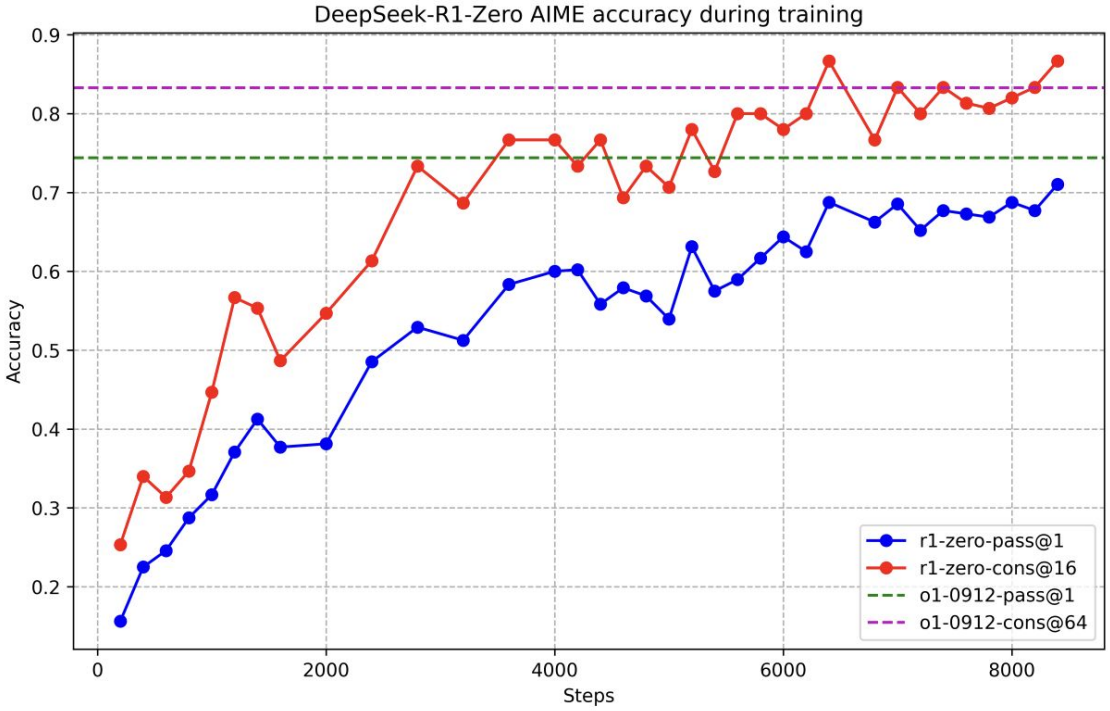

Figure 2 | AIME accuracy of DeepSeek-R1-Zero during training. For each question, we sample 16 responses and calculate the overall average accuracy to ensure a stable evaluation.

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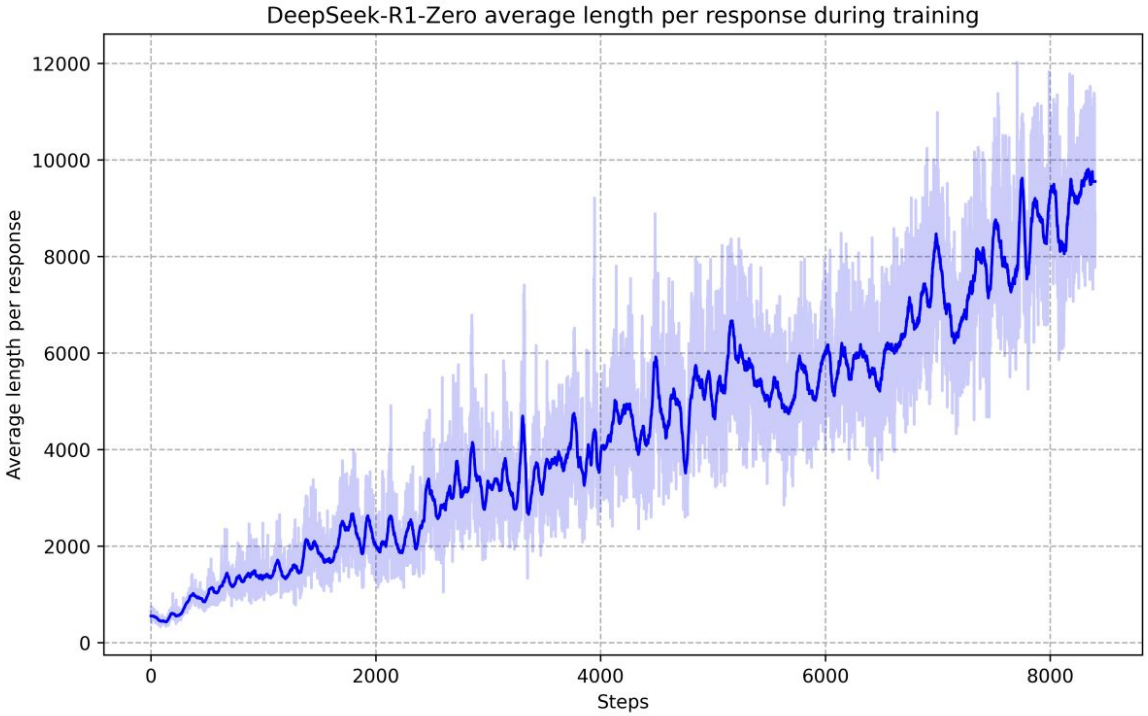

Figure 3 | The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time.

DeepSeek-AI 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. https://arxiv.org/abs/2501.12948

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to
Response: <think> To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$ $\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$ Rearrange to isolate the inner square root term: $(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$ $\ldots$ Wait, wait. Wait. That's an aha moment I can flag here. Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$ We started with the equation: $\sqrt{a - \sqrt{a + x}} = x$ First, let's square both sides: $a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$ Next, I could square both sides again, treating the equation: $\cdots$ $\ldots$

Table 3 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

DeepSeek-AI 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. https://arxiv.org/abs/2501.12948

CHAT deepseek-ai/DeepSeek-R1

UI API

Emily buys 3 apples and 2 oranges. Each orange costs $2. The total cost of all the fruit is $13. What is the cost of each apple?

<think> Okay, let me try to figure out this problem. So, Emily buys 3 apples and 2 oranges. Each orange costs $2, and the total is $13. We need to find the cost of each apple. Hmm, okay. Let's break this down step by step.

First, let's note down what we know. The problem says there are 3 apples and 2 oranges. Each orange is $2, so maybe I can calculate the total cost of the oranges first. If one orange is $2, then two oranges would be 2 times $2. Let me write that out: 2 oranges * $2/orange. That should be 2*2 = $4. So the oranges cost $4 in total.

Now, the total cost for all the fruit is $13. If the oranges are $4, then the apples must account for the remaining cost. So, subtract the cost of oranges from the total to get the cost of apples. That would be $13 total - $4 for oranges = $9 for apples. Okay, so the apples cost $9 altogether.

But Emily bought 3 apples. So if 3 apples cost $9, then each apple must cost $9 divided by 3. Let me write that: $9 / 3 apples = $3 per apple. So, each apple costs $3? Let me check that again to be sure.

Let's verify the total cost. If each apple is $3, then 3 apples would be 3*3 = $9. Then 2 oranges at $2 each would be 2*2 = $4. Adding those together, $9 + $4 = $13, which matches the total given. So that seems right. Hmm, wait, let me make sure I didn't make any mistakes in my calculations.

Starting over: 2 oranges at $2 each is 2*2=4. Total is 13. So 13-4=9 left for apples. 9 divided by 3 apples is 3. Yes, that seems correct. Alternatively, I can set up an equation. Let me try that method to confirm.

Let the cost of each apple be x dollars. Then, the total cost for apples is 3x. The total cost for oranges is 2*2=4. So the equation is 3x + 4 = 13. Solving for x: subtract 4 from both sides, 3x = 9, then divide by 3, x=3. Yep, same result. So that confirms it. Each apple costs $3. I think that's the answer. Let me just see if there's another way to approach it, but I think the equation method is straightforward.

Alternatively, maybe using ratios? Hmm, but that might complicate things. Since we have exact numbers, setting up the equation is the simplest way. Yeah, I think I'm confident with this answer. Each apple is $3. The check also works

https://api.together.ai/playground/chat/deepseek-ai/DeepSeek-R1

Group Relative Policy Optimization (GRPO)

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

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Xiao Bi[1], Haowei Zhang[1], Mingchuan Zhang[1], Y.K. Li[1], Y. Wu[1], Daya Guo[1*]

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https://github.com/deepseek-ai/DeepSeek-Math

Zhihong et al. 2024. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. https://arxiv.org/abs/2402.03300

(reminder) RL in language modeling

State S → Policy π → Distribution on Actions A

Students →

→

a	0.1%
are	3%
can	3%
…	…
…	…
the	0.5%
opened	5%
zebra	0.1%

S_0, A_0 (i.e. selecting ‘opened’),

(reminder) RL in language modeling

State S

Policy π

Distribution on
Actions A

Students opened

a	10%
an	10%
books	2%
…	…
…	…
the	10%
their	10%
zebra	0.1%

S_0, A_0, S_1, A_1, (i.e. selecting ‘their’)

(reminder) RL in language modeling

State S

Policy π

Distribution on
Actions A

Students opened their

apples	1%
ants	1%
books	10%
…	…
…	…
presents	7%
zebra	2%

Update and improve

$S_0, A_0, S_1, A_1, S_2, A_2, S_3, \ldots, A_{n_1}, S_n.$

Reward: R

(reminder) RL in language modeling

State S

Policy π

Distribution on
Actions A

Students opened their

apples	1%
ants	1%
books	10%
…	…
…	…
presents	7%
zebra	2%

$$\pi(A|S) = P(\text{Choose A when at S})$$

(reminder) RL in language modeling

State S

Policy π

Distribution on
Actions A

Students opened their

Params θ

apples	1%
ants	1%
books	10%
…	…
…	…
presents	7%
zebra	2%

$$\pi(A|S, \theta) = P(\text{Choose A when at S, params} = \theta)$$

The REINFORCE Algorithm

How to improve θ?

- Observe S_0, A_0, S_1, A_1, … , A_{n_1}, S_n and R
- If R is <u>large</u>, then update θ to

  Make actions A_t <u>more</u> likely. i.e. $\uparrow \pi(A_t|S_t, \theta)$

Large R

Students opened their books

apples	1%	
ants	1%	
books	10%	**+3%**
…	…	
…	…	
presents	7%	
zebra	2%	

The REINFORCE Algorithm

How to improve θ?

- Observe S_0, A_0, S_1, A_1, … , A_{n_1}, S_n and R
- If R is <u>large</u>, then update θ to

  Make actions A_t <u>more</u> likely. i.e. $\uparrow \pi(A_t|S_t, \theta)$

$$\mathbf{R} \ \nabla_\theta \log \pi(A_t|S_t, \theta)$$

apples	1%	
ants	1%	
books	10%	+3%
…	…	
…	…	
presents	7%	
zebra	2%	

The REINFORCE Algorithm

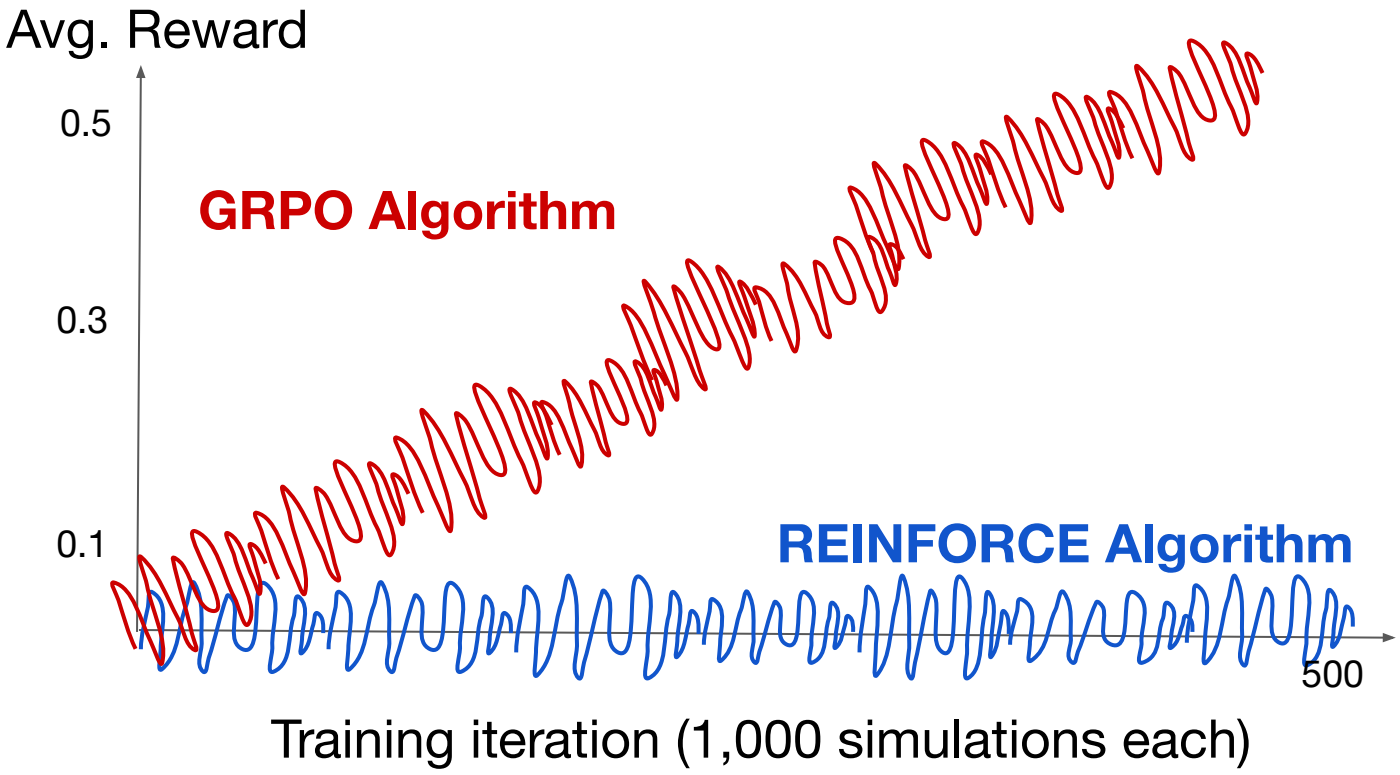
How to improve θ?

- Observe S_0, A_0, S_1, A_1, … , A_{n_1}, S_n and R
- If R is <u>large</u>, then update θ to

  Make actions A_t <u>more</u> likely. i.e. $\uparrow \pi(A_t|S_t, \theta)$

$$\theta_{new} = \theta + \mathbf{R}\ \nabla_\theta \log \pi(A_t|S_t, \theta)$$

A mockup example



Improvement: The GRPO Algorithm

How to improve θ? Reward size **R** is relative!

- Observe $S_0, A_0^{(1)}, S_1^{(1)}, A_1^{(1)}, \ldots, S_n^{(1)}$ and $R^{(1)}$
- Observe $S_0, A_0^{(2)}, S_1^{(2)}, A_1^{(2)}, \ldots, S_n^{(2)}$ and $R^{(2)}$
- Observe $S_0, A_0^{(3)}, S_1^{(3)}, A_1^{(3)}, \ldots, S_n^{(3)}$ and $R^{(3)}$

1. Students opened their books	R^1: 10
2. Students no playing	R^2: 2
3. Students were quite	R^3: 5

$$\theta_{new} = \theta + \textbf{Advantage}^{(1)} \nabla_\theta \log \pi(A_t|S_t, \theta)$$

$$\textbf{Advantage}^{(1)} = \frac{R^{(1)} - \text{avg}\{R^{(1)}, R^{(2)}, R^{(3)}\}}{\text{std}\{R^{(1)}, R^{(2)}, R^{(3)}\}}$$

DeepSeek's RL objective function

For each question q, GRPO samples a group of outputs $\{o_1, o_2, \cdots, o_G\}$ from the old policy $\pi_{\theta old}$ and then optimizes the policy model π_θ by maximizing the following objective:

Clipping: Prevent excessive jumps

KL divergence

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right),$$

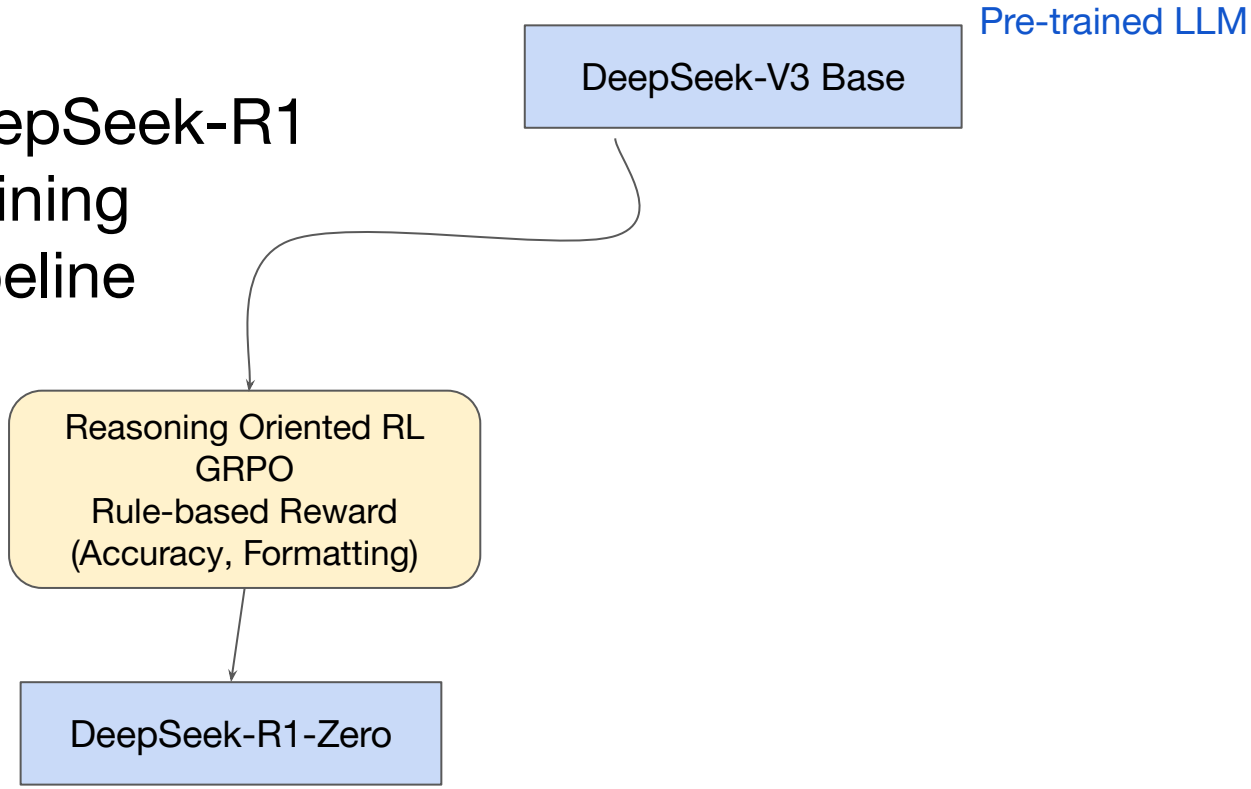
Probability Ratio

$$\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right) = \frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - 1,$$

where ε and β are hyper-parameters, and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \ldots, r_G\}$ corresponding to the outputs within each group:

$$A_i = \frac{r_i - mean(\{r_1, r_2, \cdots, r_G\})}{std(\{r_1, r_2, \cdots, r_G\})}$$

DeepSeek-R1 Training Pipeline



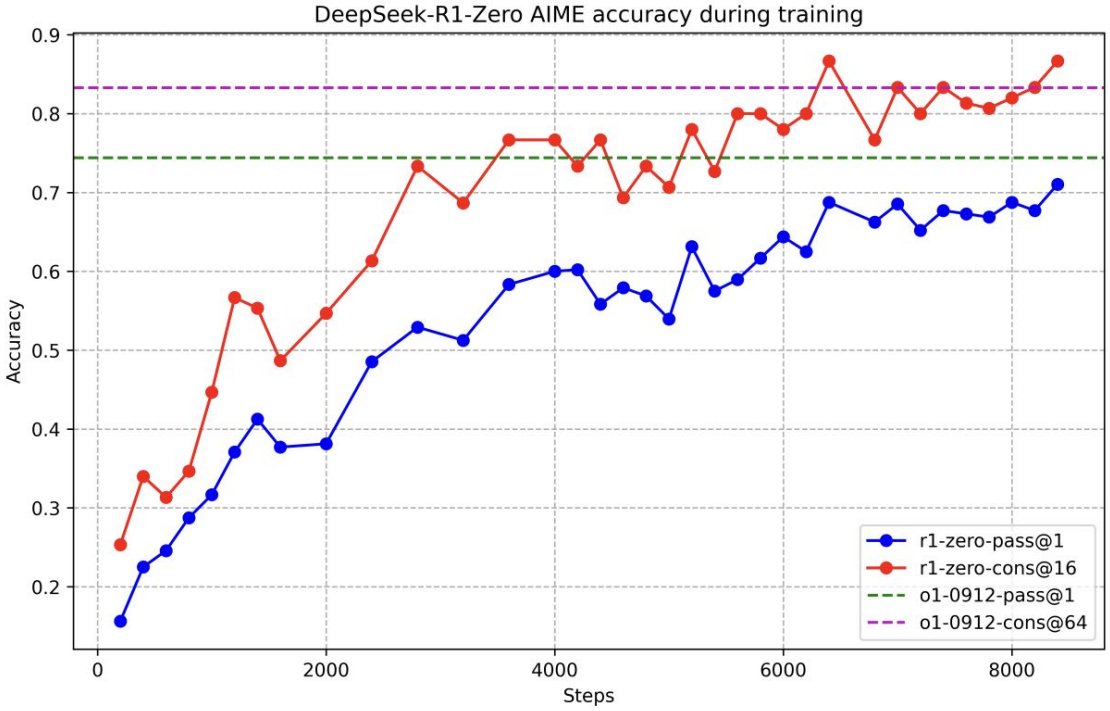
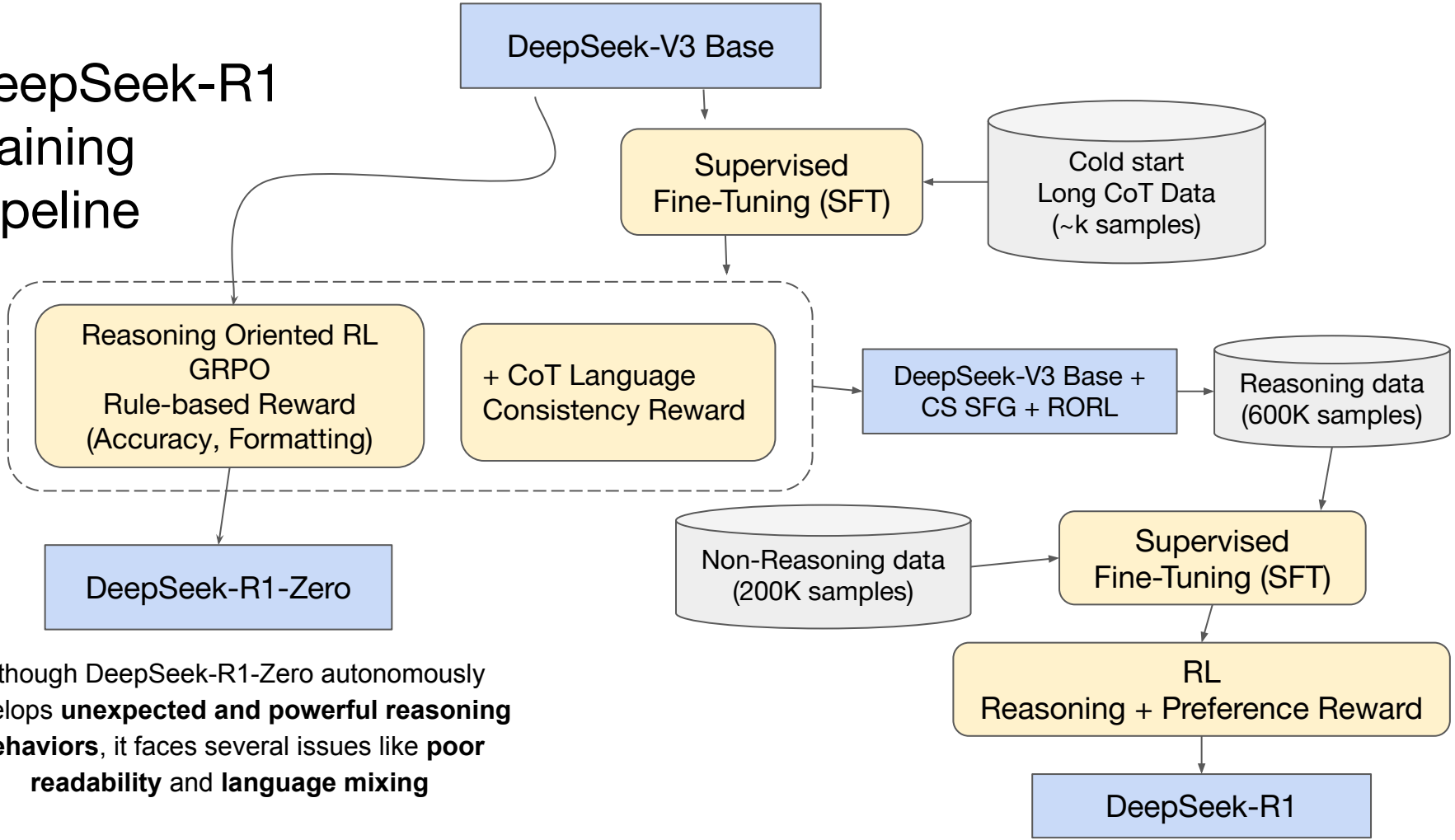


Figure 2 | AIME accuracy of DeepSeek-R1-Zero during training. For each question, we sample 16 responses and calculate the overall average accuracy to ensure a stable evaluation.

DeepSeek-AI 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. https://arxiv.org/abs/2501.12948

DeepSeek-R1 Training Pipeline



Although DeepSeek-R1-Zero autonomously develops **unexpected and powerful reasoning behaviors**, it faces several issues like **poor readability** and **language mixing**

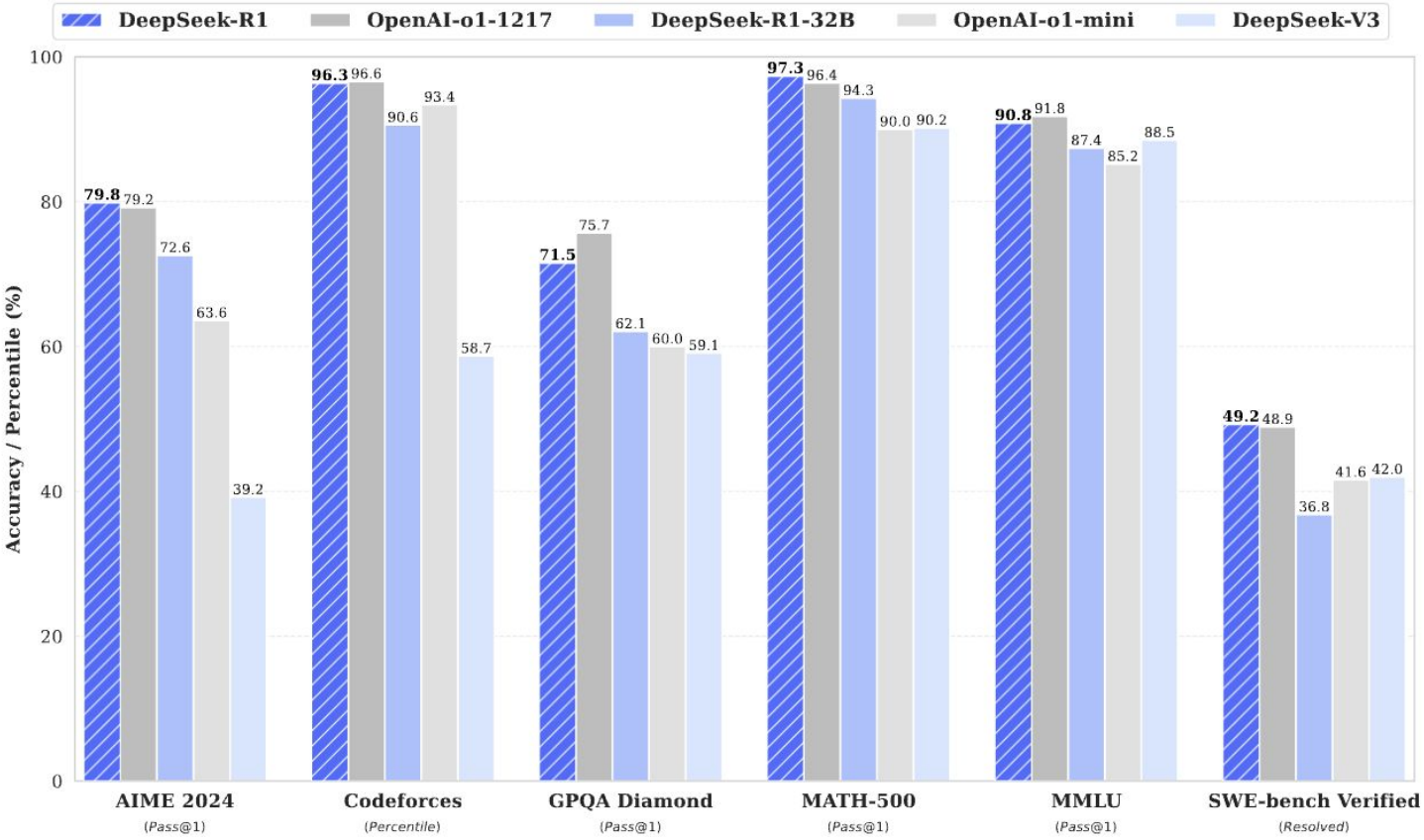
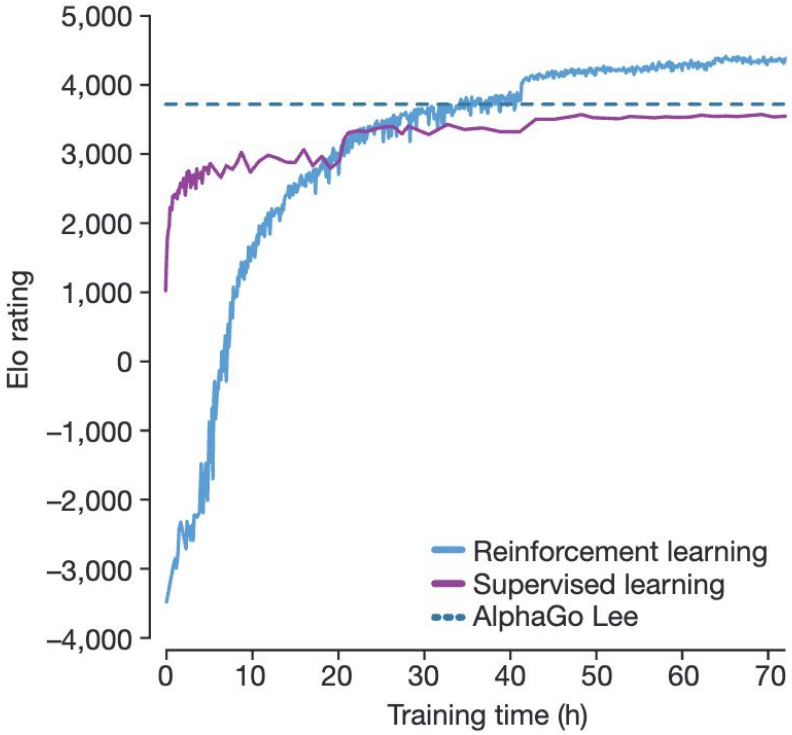


Figure 1 | Benchmark performance of DeepSeek-R1.

DeepSeek-AI 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. https://arxiv.org/abs/2501.12948

Why RL?



Silver et al. 2017. Mastering the game of go without human knowledge. Nature. https://www.nature.com/articles/nature24270

Any Questions?